One Pricer a Day

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1 Introduction

This is a project where I develop derivative pricers using **C++** and occasionally **Python**. My goal is to do at least one a day and you can access all of my source files and header files here: **GitHub Repo**

Part I
European Options

2 Closed-Form Solutions

2.1 Day 1: Black-Scholes Equation

The first option that I am going to be pricing is the European option using the closed-form Black-Scholes equation.

In **GitHub** the full **C++** script is on this link: **source file**
The **header file** (first listing) is on this link: **header file**

2.1.1 Risk-Neutral Measure

The Black-Scholes equation takes the assumption of the **risk-neutral pricing measure**

$$dS_t = rS_t dt + \sigma S_t dW_t \tag{2.1}$$

[1]
where S_t is the price of the asset at a given time t, dS_t is the infinitesimal change in the asset's price, r is the risk-free rate of interest, dt is the infinitesimal change in time, σ is the implied volatility and dW_t is the infinitesimal change in the **Wiener F function**. The **Wiener function** is defined with the property:

$$W_T - W_t \sim \mathcal{N}(0, \sqrt{T-t}) \tag{2.2}$$

where $\mathcal{N}(\mu, \sigma)$ is the **normal distribution** with mean μ and standard-deviation σ.[2]

2.1.2 Derivation of the Black-Scholes PDE

Ito's Lemma states that:

$$df(t, S_t) = \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial S_t}dS_t + \frac{1}{2}\frac{\partial^2 f}{\partial S_t^2}(dS_t)^2 \tag{2.3}$$

Substituting equation 2.1 for dS_t we get:

$$\begin{aligned} df(t, S_t) &= \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial S_t}\left(rS_t dt + \sigma S_t dW_t\right) + \frac{1}{2}\frac{\partial^2 f}{\partial S_t^2}\left(rS_t dt + \sigma S_t dW_t\right)^2 \\ &= \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial S_t}\left(rS_t dt + \sigma S_t dW_t\right) + \frac{1}{2}\frac{\partial^2 f}{\partial S_t^2}\left(r^2 S_t^2 (dt)^2 + 2r\sigma S_t^2 (dt)(dW_t) + \sigma^2 S_t^2 (dW_t)^2\right) \end{aligned} \tag{2.4}$$

The **Ito's multiplication rules** state that:

$$(dt)^2 = 0 \tag{2.5}$$
$$(dt)(dW_t) = 0 \tag{2.6}$$
$$(dW_t)^2 = dt \tag{2.7}$$

Substituting equations 2.5, 2.6, and 2.7 into equation 2.4 we get:

$$\begin{aligned} df(t, S_t) &= \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial S_t}\left(rS_t dt + \sigma S_t dW_t\right) + \frac{1}{2}\frac{\partial^2 f}{\partial S_t^2}\left(0 + 0 + \sigma^2 S_t^2 (dt)\right) \\ &= \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial S_t}\left(rS_t dt + \sigma S_t dW_t\right) + \frac{1}{2}\frac{\partial^2 f}{\partial S_t^2}\sigma^2 S_t^2 \\ &= (\frac{\partial f}{\partial t} + rS_t\frac{\partial f}{\partial S_t} + \frac{1}{2}\sigma^2 S_t^2\frac{\partial^2 f}{\partial S_t^2})dt + \sigma S_t\frac{\partial f}{\partial S_t}dW_t \end{aligned} \tag{2.8}$$

Define

$$V(t, S) \equiv f(t, S_t) \tag{2.9}$$

Using the result from equation 2.8

$$dV(t, S) = (\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S_t^2\frac{\partial^2 V}{\partial S^2})dt + \sigma S\frac{\partial V}{\partial S}dW_t \tag{2.10}$$

Now we will define a delta-hedged portfolio Π and $\Delta = \frac{\partial V}{\partial S}$ that satisfies:

$$\Pi = V - \Delta S \tag{2.11}$$

For an infinitesimal time step this becomes:

$$d\Pi = dV - \Delta dS \tag{2.12}$$

Substituting equation 2.10 for dV and equation 2.1 for dS we get.

$$\begin{aligned} d\Pi &= (\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S_t^2\frac{\partial^2 V}{\partial S^2})dt + \sigma S\frac{\partial V}{\partial S}dW_t - \Delta(rSdt + \sigma SdW_t) \\ &= (\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S_t^2\frac{\partial^2 V}{\partial S^2} - \Delta rS)dt + (\sigma S\frac{\partial V}{\partial S} - \Delta\sigma S)dW_t \end{aligned} \tag{2.13}$$

Since $\Delta = \frac{\partial V}{\partial S}$, equation 2.13 becomes:

$$\begin{aligned} d\Pi &= (\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rS\frac{\partial^2 V}{\partial S^2})dt + (\sigma S\frac{\partial V}{\partial S} - \sigma S\frac{\partial V}{\partial S})dW_t \\ &= (\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rS\frac{\partial V}{\partial S})dt \\ &= (\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2})dt \end{aligned} \tag{2.14}$$

The **arbitrage condition** in a **risk-free portfolio** is:

$$d\Pi = r\Pi dt \tag{2.15}$$

Substituting this into equation 2.14 we get:

$$\begin{aligned} r\Pi dt &= (\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2})dt \\ r\Pi &= \frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \end{aligned} \tag{2.16}$$

Substituting equation 2.11 into equation 2.16 we get:

$$\begin{aligned} r(V - \Delta S) &= \frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \\ r(V - \frac{\partial V}{\partial S}S) &= \frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \\ rV - rS\frac{\partial V}{\partial S} &= \frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \end{aligned} \tag{2.17}$$

This gives us the final partial differential equation which is also the **Black-Scholes equation**:

$$\boxed{\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0} \tag{2.18}$$

In this equation V is the option's price, t is the current time.
$\frac{\partial V}{\partial t}$ is the rate of change of the option's price to time, $\frac{\partial V}{\partial S}$ is the rate of change of the option's price to the underlying stock price and $\frac{\partial^2 V}{\partial S^2}$ is a concavity factor on how the option price changes with the stock price. The Black-Scholes can be solved for a call option and a put option. [3].

2.1.3 Closed-Form Solution for European Call Options

The closed-form solution for the Black-Scholes equation where the option is a call option is:

$$\begin{aligned} C(t,S) &= S\Phi(d_1) - Ke^{-r(T-t)}\Phi(d_2) \\ d_1 &= \frac{\ln(\frac{S}{K}) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}} \\ d_2 &= d_1 - \sigma\sqrt{T-t} \end{aligned} \tag{2.19}$$

In this equation $T-t$ represents the time to maturity and K represents the strike price. [3].

$\Phi(z)$ is the standard normal cumulative distribution function which can be represented with the error function $\text{erf}(z)$:

$$\Phi(z) = \frac{1}{2}(1 + \text{erf}(\frac{z}{\sqrt{2}})) \tag{2.20}$$

In C++ this can be implemented as:

```
1  #include <iostream>
2  #include <cmath>
3  #include <type_traits>
4
5  template <typename T>
6  auto normalCDF(T x) -> std::common_type_t <double , T>
7  {
8          using commonType = std::common_type_t<double, T>;
9          return static_cast<commonType>(0.5) * (static_cast<
               commonType>(1) + std::erf(x / std::sqrt(2.0)));
10 }
```

The function we are creating is called **normalCDF**. This listing uses the **cmath** header file in order to import **std::erf**. It also uses a function template with a type placeholder **T** so that we can use this function for any type given. In order to ensure that this function has at least a type **double** we use the header file **type traits** which allows us to specify that we want this function to be at least double in line 6 trailing return type. In line 8 we ensure that the **commonType** alias used amongst all the literals and variables in this function are also at least of type **double**. This function returns the result shown in equation 2.20 with a type of at least **double**.

Now to implement the function shown in equation 2.19 in C++ we use:

```
1  #include "functions.hpp" // header file for normalCDF
2
3  template <typename SType, typename KType, typename rType, typename
       sigmaType, typename tType>
4  auto blackScholesCall(SType S, KType K, rType r, sigmaType sigma,
       tType t) -> std::common_type_t<double, SType, KType, rType,
       sigmaType, tType>
5  {
6          using commonType = std::common_type_t<double, SType, KType,
               rType, sigmaType, tType>;
7          auto S_new {static_cast<commonType>(S)};
8          auto K_new {static_cast<commonType>(K)};
9          auto r_new {static_cast<commonType>(r)};
10         auto sigma_new {static_cast<commonType>(sigma)};
11         auto t_new {static_cast<commonType>(t)};
12         auto sqrt_t_new {std::sqrt(t_new)};
13
14         auto d1 {(std::log(S_new / K_new) + (r_new + (static_cast<
               commonType>(0.5) * sigma_new * sigma_new) * t_new) / (
               sigma_new * sqrt_t_new)};
15         auto d2 {d1 - sigma_new * sqrt_t_new};
16
17         auto C {S_new * normalCDF(d1) - K_new * std::exp(-r_new *
               t_new) * normalCDF(d2)};
18         return C;
19 }
```

The function in this listing is called **blackScholesCall**. This listing again uses the same header files and the same principles as the one before and converts every thing into a **commonType** of at least type **double**. The formula for d_1 is shown in line 12 and the formula for d_2 is shown in line 13. Finally the result in equation 2.19 is shown in line 16 using the previously defined function **normalCDF**. The function returns the value of the **call option** with at least type **double**.

2.1.4 Closed-Form Solution for European Put Options

The closed-form solution for the Black-Scholes equation fora put option is:

$$P(t, S) = Ke^{-r(T-t)}\Phi(-d_2) - S\Phi(-d_1) \tag{2.21}$$

using the same definitions of d_1 and d_2 in 2.19 [3]

The implementation of this in C++ is:

```
1  #include "functions.hpp" // header file for normalCDF
2
3  template <typename SType, typename KType, typename rType, typename
       sigmaType, typename tType>
4  auto blackScholesPut(SType S, KType K, rType r, sigmaType sigma,
       tType t) -> std::common_type_t<double, SType, KType, rType,
       sigmaType, tType>
5  {
6          using commonType = std::common_type_t<double, SType, KType,
               rType, sigmaType, tType>;
7          auto S_new {static_cast<commonType>(S)};
8          auto K_new {static_cast<commonType>(K)};
9          auto r_new {static_cast<commonType>(r)};
10         auto sigma_new {static_cast<commonType>(sigma)};
11         auto t_new {static_cast<commonType>(t)};
12         auto sqrt_t_new {std::sqrt(t_new)};
13
14         auto d1 {(std::log(S_new / K_new) + (r_new + (static_cast<
               commonType>(0.5) * sigma_new * sigma_new) * sqrt_t_new) /
                (sigma_new * sqrt_t_new)};
15         auto d2 {d1 - sigma_new * std::sqrt(t_new)};
16
17         auto P {K_new * std::exp(-r_new * t_new) * normalCDF(-d2) -
               S_new * normalCDF(-d1)};
18         return P;
19 }
```

The function in this listing is called **blackScholesPut**. This listing is almost identical to the one for **blackScholesCall**. The formula for d_1 is again shown in line 12 and the formula for d_2 is shown in line 13. Finally the result in equation 2.21 is computed in line 16 (stored as **P**) using the function **normalCDF**. The function returns the value of the **put option** with at least type **double**.

2.2 Day 2: Black-76 PDE

The Black-76 model is just an adaption of the **Black-Scholes equation** except it uses the price of futures and bonds to price options.

In **GitHub** the source file can be found here: **source file**

For the listings in this section I will continue using the header file **functions.hpp** that I defined on day 1. I will continue to use the **normalCDF** function from there.

2.2.1 Derivation of the Black-76 PDE

The modelling assumptions are that the futures of forward price process is modelled by **geometric Brownian motion** shown by the equation:

$$dF_t = \sigma F_t dW_t \tag{2.22}$$

where F_t is the price of a future or a forward at a given time, dF_t is the infinitesimal change of the price of a future of a forward, σ is implied volatility and dW_t is the infinitesimal change in the **Wiener function**.[4]

Substituting equation 2.22 into **Ito's Lemma** 2.3

$$\begin{aligned} df(t, S_t) &= \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial F_t}(\sigma F_t dW_t) + \frac{1}{2}\frac{\partial^2 f}{\partial F_t^2}(\sigma F_t dW_t)^2 \\ &= \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial F_t}(\sigma F_t dW_t) + \frac{1}{2}\frac{\partial^2 f}{\partial F_t^2}(\sigma^2 F_t^2 (dW_t)^2) \end{aligned} \tag{2.23}$$

Substituting equations 2.7 into equation 2.23 we get:

$$\begin{aligned} df(t, S_t) &= \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial F_t}(\sigma F_t)dW_t + \frac{1}{2}\frac{\partial^2 f}{\partial F_t^2}(\sigma^2 F_t^2)dt \\ &= (\frac{\partial f}{\partial t} + \frac{1}{2}\sigma^2 F_t^2 \frac{\partial^2 f}{\partial F_t^2})dt + \sigma F_t \frac{\partial f}{\partial F_t}dW_t \end{aligned} \tag{2.24}$$

Define

$$V(t, S) \equiv f(t, S_t) \tag{2.25}$$

Using the result from equation 2.24

$$dV(t, S) = (\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial F^2})dt + \sigma F \frac{\partial V}{\partial F}dW_t \tag{2.26}$$

The thing that is different about the **delta-hedged portfolio** is that at the time the option is being exchanged, the cost of the **future** is 0.

$$\Pi = V \tag{2.27}$$

For an infinitesimal time step, the value of the future increases by dF which leads to

$$d\Pi = dV - \Delta dF \tag{2.28}$$

where $\Delta = \frac{\partial V}{\partial F}$ Substituting equation 2.26 for dV and equation 2.22 for dF we get.

$$\begin{aligned} d\Pi &= (\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 F^2 \frac{\partial^2 V}{\partial F^2})dt + \sigma F \frac{\partial V}{\partial F} dW_t - \Delta(\sigma F dW_t) \\ &= (\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 F^2 \frac{\partial^2 V}{\partial F^2})dt + (\sigma F \frac{\partial V}{\partial F} - \Delta\sigma F)dW_t \end{aligned} \tag{2.29}$$

Since $\Delta = \frac{\partial V}{\partial S}$, equation 2.29 becomes:

$$\begin{aligned} d\Pi &= (\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 F^2 \frac{\partial^2 V}{\partial F^2})dt + (\sigma F \frac{\partial V}{\partial F} - \sigma F \frac{\partial V}{\partial F})dW_t \\ &= (\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 F^2 \frac{\partial^2 V}{\partial F^2})dt \end{aligned} \tag{2.30}$$

The **arbitrage condition** in a **risk-free portfolio** is:

$$d\Pi = r\Pi dt \tag{2.31}$$

Substituting this into equation 2.30 we get:

$$\begin{aligned} r\Pi dt &= (\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 F^2 \frac{\partial^2 V}{\partial F^2})dt \\ r\Pi &= \frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 F^2 \frac{\partial^2 V}{\partial F^2} \end{aligned} \tag{2.32}$$

Substituting equation 2.27 into equation 2.32 we get:

$$rV = \frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 F^2 \frac{\partial^2 V}{\partial F^2} \tag{2.33}$$

Which finally leads to the equation:

$$\boxed{\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 F^2 \frac{\partial^2 V}{\partial F^2} - rV = 0} \tag{2.34}$$

Which is also known as the **Black-76 Equation**

2.2.2 Black-76 Formula for Call Options

The Black-76 Formula for Call options is:

$$\begin{aligned} C(t,F) &= e^{-r(T-t)}(F\Phi(d_1) - K\Phi(d_2)) \\ d_1 &= \frac{\ln(\frac{F}{K}) + \frac{1}{2}\sigma^2(T-t)}{\sigma\sqrt{T-t}} \\ d_2 &= d_1 - \sigma\sqrt{T-t} \end{aligned} \tag{2.35}$$

[4] The implementation in C++ is:

```
1  #include "functions.hpp" // header file for normalCDF
2
3  template <typename FType, typename KType, typename rType, typename
       sigmaType, typename tType>
4  auto black76Call(FType F, KType K, rType r, sigmaType sigma, tType t
       ) -> std::common_type_t<double, FType, KType, rType, sigmaType,
       tType>
5  {
6          using commonType = std::common_type_t<double, FType, KType,
               rType, sigmaType, tType>;
7          auto F_new {static_cast<commonType>(F)};
8          auto K_new {static_cast<commonType>(K)};
9          auto r_new {static_cast<commonType>(r)};
10         auto sigma_new {static_cast<commonType>(sigma)};
11         auto t_new {static_cast<commonType>(t)};
12         auto sqrt_t_new {std::sqrt(t_new)};
13
14         auto d1 {(std::log(F_new / K_new) + (static_cast<commonType
               >(0.5) * sigma_new * sigma_new) * t_new ) / (sigma_new *
               sqrt_t_new)};
15         auto d2 {d1 - sigma_new * sqrt_t_new};
16
17         auto C {(std::exp(-r_new * t_new)) * (F_new * normalCDF(d1)
               - K_new * normalCDF(d2))};
18         return C;
19 }
```

d_1 is calculated in line 13, d_2 is calculated in line 14 and finally equation 2.35 is computed in line 16 (stored as **C**) and returned in line 17.

2.2.3 Black-76 Formula for Put Options

The Black-76 Formula for Call options is:

$$\begin{aligned} P(t,F) &= e^{-r(T-t)}(K\Phi(-d_2) - F\Phi(-d_1)) \\ d_1 &= \frac{\ln(\frac{F}{K}) + \frac{1}{2}\sigma^2(T-t)}{\sigma\sqrt{T-t}} \\ d_2 &= d_1 - \sigma\sqrt{T-t} \end{aligned} \tag{2.36}$$

[4] The implementation in C++ is:

```
1  #include "functions.hpp" // header file for normalCDF
2
3  template <typename FType, typename KType, typename rType, typename
       sigmaType, typename tType>
4  auto black76Put(FType F, KType K, rType r, sigmaType sigma, tType t)
       -> std::common_type_t<double, FType, KType, rType, sigmaType,
       tType>
5  {
6          using commonType = std::common_type_t<double, FType, KType,
               rType, sigmaType, tType>;
7          auto F_new {static_cast<commonType>(F)};
8          auto K_new {static_cast<commonType>(K)};
9          auto r_new {static_cast<commonType>(r)};
10         auto sigma_new {static_cast<commonType>(sigma)};
11         auto t_new {static_cast<commonType>(t)};
12         auto sqrt_t_new {std::sqrt(t_new)};
13
14         auto d1 {(std::log(F_new / K_new) + (static_cast<commonType
               >(0.5) * sigma_new * sigma_new) * t_new ) / (sigma_new *
               sqrt_t_new)};
15         auto d2 {d1 - sigma_new * sqrt_t_new};
16
17         auto P {std::exp(-r_new * t_new) * (K_new * normalCDF(-d2) -
               (F_new * normalCDF(-d1)))};
18         return P;
19 }
```

d_1 is calculated in line 13, d_2 is calculated in line 14 and finally equation 2.36 is computed in line 16 (stored as **P**) and returned in line 17.

2.3 Day 3: Merton Dividend-Adjusted Black-Scholes

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References

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